

```
%oldchk=PbRhaC12C12.chk
%nprocshared=64
%mem=320GB
%chk=PbRhaC12C12T2.chk
# freq apfd/gen scrf=(solvent=water) opt=loose pseudo=read scf=xqc
```

PbRhaC12C12T21

```
0 1
Pb      -0.26274800    1.20737400   -0.87859800
O        1.39563100   -0.61839100    1.69393600
O        0.67766400    2.12742600   -2.92718500
O       -0.35889000   -1.27659600   -0.13438200
C        1.80917800   -4.29492000    0.36252200
O        2.99558100   -3.49633600    0.23012000
C        3.00626000   -2.34685400    1.05019300
C        1.93466400   -1.33292100    0.58462200
C        0.81532700   -2.08503700   -0.12449800
C        0.56979200   -3.41491700    0.56896400
O        1.51939300    0.24463900   -2.14323900
C        1.55607400    1.22074400   -2.95221200
C        2.72154100    1.31233900   -3.90533500
C        3.87159300    2.11491500   -3.27928400
O        4.13898800    1.63824100   -1.93912900
C        3.59011700    3.59681300   -3.15124500
C        4.77733300    4.38809000   -2.62048400
C        4.48762800    5.87980700   -2.52473400
C        5.66860100    6.68549100   -2.00067800
C        5.38273800    8.17817300   -1.90481100
C        4.75507800    0.46030600   -1.81442900
O        5.15197300   -0.19785500   -2.74962800
C        4.88269000    0.06087300   -0.36767400
C        4.84285200   -1.46480600   -0.20738500
O        4.29210000   -1.81510100    1.07478000
C        1.95755000   -5.33367900    1.45700000
O       -0.58132200   -4.05996700    0.05777100
C        6.22639700   -2.07954900   -0.31904300
C        6.21617400   -3.59816500   -0.37834700
C        7.61464700   -4.19667400   -0.43458200
C        7.60745500   -5.71837000   -0.49389700
C        9.00167900   -6.32852900   -0.54561100
O       -0.96857300   -0.59047500    4.46215900
O       -2.82409500    1.69492000   -1.78311100
O        0.80937800   -2.67870100    3.72210400
C       -2.03531300   -4.39450000    3.48089800
O       -1.90588600   -3.44662000    2.41936200
C       -1.91144400   -2.09620400    2.83115300
C       -1.43237100   -1.91324300    4.26944700
C       -0.30514200   -2.87306800    4.57551400
C       -0.81952000   -4.31913400    4.41332000
```

O	-1.59862000	-0.09098800	-2.24846200
C	-2.69534900	0.57001700	-2.29902500
C	-3.85348400	-0.09032500	-3.01822000
C	-5.15117800	0.02822200	-2.24542200
O	-4.91559300	-0.62709000	-0.97010200
C	-6.30872000	-0.64513900	-2.96005300
C	-7.63499900	-0.56178600	-2.21858900
C	-8.77274500	-1.22199200	-2.98554100
C	-10.10351000	-1.15588900	-2.24833600
C	-11.24674300	-1.81379400	-3.00930200
C	-5.37153100	-0.04168700	0.14297900
O	-5.99123700	0.99564200	0.17113700
C	-4.97349700	-0.84030500	1.35857500
C	-3.45721300	-0.87463100	1.51402000
O	-3.19004200	-1.53398300	2.75560300
C	-2.20216400	-5.75854100	2.85882600
O	-1.25996800	-4.86951800	5.64063700
C	-2.81912200	0.50908500	1.50239800
C	-3.33559300	1.47446900	2.56025200
C	-2.47654100	2.72888400	2.64234800
C	-2.95948200	3.72319700	3.68822600
C	-2.09427000	4.97379800	3.76627100
H	-3.03930400	-1.47375800	0.69809800
H	-5.37498200	1.07564300	-2.03955600
H	4.20634600	-1.88789000	-0.98924200
H	4.77094000	1.94869100	-3.87854100
H	1.70139600	-4.78406200	-0.61078400
H	2.78800200	-2.61458600	2.09086100
H	2.36832700	-0.62739000	-0.12556500
H	1.13478100	-2.26188100	-1.15683100
H	0.43026500	-3.23526400	1.63643100
H	3.06638800	0.30574100	-4.13943700
H	2.41444400	1.81335500	-4.82474500
H	3.31094900	3.96803200	-4.14347900
H	2.71473800	3.74326800	-2.50947900
H	5.06092700	4.00663400	-1.63334700
H	5.64476800	4.22498000	-3.27339200
H	4.19751000	6.25971500	-3.51328400
H	3.61873100	6.04033600	-1.87288700
H	5.95780200	6.30567600	-1.01173200
H	6.53748700	6.52253200	-2.65211700
H	4.51421300	8.34116400	-1.25280600
H	5.09209600	8.55676500	-2.89386100
H	5.82902700	0.46341900	0.01189400
H	4.09200300	0.54430200	0.20547800
H	1.05300800	-5.94494000	1.50792600
H	2.09785000	-4.85615500	2.42991900
H	2.80677900	-5.99061000	1.25808800
H	-1.25597200	-3.96201800	0.74905500
H	6.70164500	-1.66930600	-1.21636600

H	6.82038900	-1.73585600	0.53822100
H	5.67848500	-3.99095700	0.49045400
H	5.64300700	-3.92168000	-1.25641500
H	8.15048800	-3.80028000	-1.30753200
H	8.18703800	-3.86985600	0.44397700
H	7.07004800	-6.11255200	0.37897600
H	7.03468200	-6.04486400	-1.37222100
H	9.57453300	-6.00172100	0.33255900
H	9.53895000	-5.93532200	-1.41901500
H	-2.92662300	-4.14695800	4.07151700
H	-1.22818000	-1.58245100	2.15027800
H	-2.26283000	-2.13209900	4.95282200
H	0.00042600	-2.73425300	5.62104000
H	0.00221100	-4.90985100	3.99026000
H	-3.97508600	0.40824700	-3.98541400
H	-3.61366300	-1.13735300	-3.21263100
H	-6.40428900	-0.17095900	-3.94384000
H	-6.04804600	-1.69455600	-3.14486400
H	-7.54009200	-1.03959000	-1.23661700
H	-7.88100200	0.48851800	-2.02369400
H	-8.87613300	-0.74445100	-3.96884400
H	-8.51797900	-2.27138800	-3.18459000
H	-9.99822200	-1.63365700	-1.26519600
H	-10.35608900	-0.10622200	-2.04774000
H	-10.99342500	-2.86329500	-3.21031400
H	-11.35212200	-1.33569100	-3.99231500
H	-5.43401700	-0.38360700	2.23465400
H	-5.34860900	-1.86205600	1.25904300
H	-2.25896700	-6.51625600	3.64238500
H	-1.34989800	-5.99141900	2.21536400
H	-3.11490300	-5.80590800	2.26149800
H	-0.49264700	-5.03616800	6.19723400
H	-1.73848300	0.37561400	1.63409700
H	-2.96271100	0.95092900	0.51010600
H	-4.37028100	1.75547200	2.33633300
H	-3.37201900	0.97995100	3.54008000
H	-1.43754500	2.44625900	2.85971500
H	-2.45420700	3.21495000	1.65811400
H	-3.99614700	4.00945200	3.46740200
H	-2.98352500	3.23398800	4.67107000
H	-2.07043700	5.46088500	2.78234400
H	-1.05747400	4.68514600	3.98481900
H	-1.69114400	0.01257400	4.25390000
H	0.85853400	-1.75088000	3.44876100
H	-0.96988600	-1.65859700	-0.77943700
H	2.06874500	-0.00597300	2.00960300
C	-12.57727200	-1.74839200	-2.27120400
H	-12.83016600	-0.69877200	-2.07005900
H	-12.47151700	-2.22633200	-1.28809100
C	-13.72134500	-2.40604500	-3.03114200

H	-13.82777400	-1.92861000	-4.01456600
H	-13.46943700	-3.45603600	-3.23228600
C	-15.05287600	-2.34061900	-2.29348300
H	-15.30288800	-1.29113000	-2.09315900
H	-14.94477900	-2.81772700	-1.31126400
C	-16.18851700	-3.00187900	-3.06441600
H	-17.13589700	-2.94535600	-2.52099900
H	-16.33473600	-2.52202300	-4.03738500
H	-15.97451900	-4.05935500	-3.24950600
C	-2.57123200	5.97234300	4.81232200
H	-3.60817900	6.26051100	4.59388700
H	-2.59482300	5.48493900	5.79616400
C	-1.70560800	7.22281800	4.88929100
H	-1.68219400	7.71101400	3.90567900
H	-0.66818200	6.93514100	5.10701000
C	-2.18058500	8.22282000	5.93600600
H	-3.21714100	8.50878000	5.71732600
H	-2.20338600	7.73352900	6.91790400
C	-1.30617600	9.46874700	6.00211700
H	-1.29284400	9.99105100	5.04017500
H	-1.66094100	10.17488700	6.75809800
H	-0.27149600	9.21015900	6.24886100
C	6.56402200	8.98544800	-1.38315000
H	7.43302200	8.82001300	-2.03399200
H	6.85329900	8.60853600	-0.39300900
C	6.27998000	10.47870000	-1.29135000
H	5.99079000	10.85616500	-2.28144500
H	5.41107000	10.64531200	-0.64048300
C	7.46133900	11.28745500	-0.77018100
H	8.32868400	11.11898500	-1.42088200
H	7.74858400	10.90965200	0.21917300
C	7.16425700	12.77934800	-0.68492500
H	8.02203800	13.34396000	-0.30883300
H	6.90479100	13.18522400	-1.66790300
H	6.31951100	12.97437500	-0.01658200
C	8.99086400	-7.85037600	-0.60289300
H	8.45221600	-8.24299600	0.26997600
H	8.41860800	-8.17696900	-1.48154700
C	10.38425700	-8.46266200	-0.65256400
H	10.92374600	-8.07105700	-1.52549500
H	10.95722200	-8.13682000	0.22605200
C	10.37317300	-9.98516400	-0.70945400
H	9.83334100	-10.37457300	0.16292000
H	9.80051600	-10.30896100	-1.58768600
C	11.77287400	-10.58496300	-0.75813200
H	11.74596900	-11.67750700	-0.79882700
H	12.32081500	-10.23331600	-1.63814400
H	12.35359400	-10.29980200	0.12491500

C H O 0

6-311+g(2d,p)

Pb	0	
S	4	1.00
	591.61124370	0.22126521076D-03
	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2	1.00
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1	1.00
	1.6600600927	1.0000000
S	1	1.00
	0.80431655001	1.0000000
S	1	1.00
	0.22627039020	1.0000000
S	1	1.00
	0.84014530665D-01	1.0000000
P	3	1.00
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3	1.00
	2.2021426123	0.40196284806
	1.2209125119	0.46058131862
	0.63367559815	0.19367655397
P	1	1.00
	0.28202837058	1.0000000
P	1	1.00
	0.11333375666	1.0000000
P	1	1.00
	0.43948707430D-01	1.0000000
D	6	1.00
	61.315369628	0.33870800787D-03
	12.372195840	0.13788683942D-01
	6.9254944983	-0.75979608103D-01
	2.3319539939	0.28113784298
	1.2108730003	0.44474512269
	0.60090478506	0.35326874351
D	1	1.00
	0.28135869813	1.0000000
D	1	1.00
	0.11500000000	1.0000000
F	1	1.00
	1.0000000	1.0000000
F	1	1.00
	0.2896200	1.0000000

PB	0	
PB-ECP	3	60
f potential		
2		
2	3.88751200	12.20989200
2	3.81196300	16.19029100
s-f potential		
4		
2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
p-f potential		
6		
2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
d-f potential		
6		
2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600
2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100